

ANDROID DEVELOPER VIRTUAL INTERNSHIP

An Internship Report Submitted at the end of the Seventh semester

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

(Data Science)

Submitted By
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(223J1A44C3)

Under the esteemed Guidance of

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)



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2025-2026

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(DATA SCIENCE)

CERTIFICATE

This is to certify that this Internship entitled “**ANDROID DEVELOPER VIRTUAL INTERNSHIP**” done by “**NALLIMILLI SARANYA REDDY (223J1A44C3)**” is a student of B.Tech in the Department of Computer Science and Engineering – Data Science, Raghu Institute of Technology, during the period 2022-2026, in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science and Engineering(Data Science) to the Jawaharlal Nehru Technological University, Gurajada Vizianagaram is a record of bonafide work carried out under my guidance and supervision.

The results embodied in this internship report have not been submitted to any other University or Institute for the award of any Degree

Internal Guide

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DISSERTATION APPROVAL SHEET

This is to certify that the dissertation

ANDROID DEVELOPER

BY

Nallimili Saranya Reddy

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*is approved of the degree of **Bachelor of Technology***

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(Assistant Professor)

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DECLARATION

This is to certify that this internship titled “**ANDROID DEVELOPER VIRTUAL INTERNSHIP**” is bonafide work done by me, in partial fulfillment of the requirements for the award of the degree B.Tech, and submitted to the **Department of Computer Science and Engineering(Data Science), Raghu Institute of Technology, Dakamari.**

I also declare that this internship is a result of my effort and that has not been copied from anyone and I have taken only citations from the sources which are mentioned in the references.

This work was not submitted earlier at any other University or Institute for the reward of any degree.

Date:

Place:

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Certificate of Virtual Internship

This is to certify that

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Android Developer Virtual Internship

During April - June 2025

We wish him / her all the best for the future endeavours

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Regards

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ABSTRACT

In today's dynamic tech landscape, hands-on experience is indispensable for aspiring developers. The Google Android Developer Virtual Internship stands out as a pioneering initiative aimed at equipping participants with practical skills in Android app development. This abstract delves into the structure and objectives of the internship program, offering a comprehensive overview of its curriculum and learning outcomes.

Android app development using Kotlin has witnessed remarkable growth and adoption since

Google endorsed it as an official language in 2017. Statistics reveal that over 80% of the top 1000 Android apps on the Google Play Store are leveraging Kotlin for development, indicating a strong preference among developers. This surge in adoption is attributed to Kotlin's modern language features, improved syntax, and enhanced productivity compared to Java. Developers praise Kotlin for its concise syntax, null safety, extension functions, and support for coroutines, which facilitate asynchronous programming. Moreover, Kotlin's seamless interoperability with Java enables developers to gradually transition existing codebases to Kotlin, ensuring a smooth migration process.

Performance-wise, Kotlin offers tangible benefits over Java, with reduced boilerplate code and increased developer efficiency. Feedback from developers consistently highlights the language's expressiveness, safety features, and ease of integration with existing Android tools and libraries. The satisfaction level among developers using Kotlin for Android development remains high, supported by its robust community and ongoing support from Google and JetBrains, the language's creators.

Through a blend of guided tutorials, coding challenges, and real-world projects, interns engage in the complete app development lifecycle, from conceptualization to deployment. The abstract highlights the program's emphasis on fostering collaboration, innovation, and problemsolving abilities within a virtual environment. Drawing on participant testimonials and performance evaluations, it evaluates the effectiveness of the internship in enhancing technical proficiency and preparing individuals for careers in mobile app development. Furthermore, it discusses the scalability and potential impact of virtual internships in democratizing access to practical learning opportunities. By shedding light on the Google Android Developer Virtual Internship, this abstract contributes to the discourse on innovative approaches to skill development in the digital age.

INTRODUCTION

The Google Android Development Virtual Internship stands as a testament to the evolving landscape of technology education and workforce development. This innovative program represents a collaborative effort between Google, a global leader in technology, and aspiring developers worldwide seeking to gain practical experience in Android app development. Through a blend of interactive learning modules, real-world projects, and mentorship opportunities, the internship aims to equip participants with the skills and knowledge necessary to thrive in the competitive field of mobile app development.

In this introduction, we delve into the structure, objectives, and significance of the Google Android Development Virtual Internship, shedding light on its role in shaping the next generation of Android developers. We explore how the program leverages Google's expertise in Android development to provide participants with a comprehensive learning experience, fostering collaboration, innovation, and professional growth within a virtual environment.

The introduction sets the stage for understanding the transformative impact of the Google Android Development Virtual Internship, highlighting its contributions to skill development, industry readiness, and the broader ecosystem of technology education. Through an exploration of its key components and underlying principles, we aim to provide a comprehensive overview of this ground-breaking initiative and its implications for the future of Android app development.

Moreover, the Google Android Development Virtual Internship serves as a bridge between theoretical knowledge and real-world application, offering participants the opportunity to apply their skills in practical scenarios. By working on authentic projects that address industry challenges and requirements, interns gain invaluable insights into the app development lifecycle, including ideation, design, implementation, and deployment. This hands-on experience not only enhances technical proficiency but also cultivates essential soft skills such as problem-solving, teamwork, and communication, which are vital for success in the technology industry. Through this immersive learning journey, participants emerge not only as proficient Android developers but also as adaptable, creative problem solvers prepared to tackle the complexities of modern software development.

TABLE OF CONTENTS

S.NO.	MODULE	CONTENT	PAGE NO.
1.	Module 1	Your First Android App	10
2.	Module 2	Building App UI	12
3.	Module 3	Display Lists And Use Material Design	13
4.	Module 4	Navigation And App Architecture	14
5.	Module 5	Connect To The Internet	15
6.	Module 6	Data Persistence	17
7.	Module 7	Work Manager	18
8.	Module 8	Example Projects	19
9.	Reports		23
10.	Conclusion		25

MODULE 1 : YOUR FIRST ANDROID APP

Introduction

This module introduces the basic setup and structure of an Android project. As a beginner, the goal is to become familiar with Android Studio, understand what files make up an app, and successfully run your first simple program — the “Hello World” app.

Android apps are primarily built using **Kotlin** or **Java**, and they run on devices with Android OS. Android Studio is a comprehensive **Integrated Development Environment (IDE)** that offers an editor, debugging tools, and an emulator to test your apps.

Setting up Android Studio :

ANDROID STUDIO



Android Studio is the official IDE for Android development. It provides tools for writing code, designing interfaces, debugging, and testing.

Steps:

1. Download and install Android Studio.
2. Create a new project → Select **Empty Activity**.
3. Choose a name (for example, “MyFirstApp”).
4. Select **Kotlin** as the programming language.
5. Click **Finish** to create your first project.

Understanding the Project Structure

Each Android app consists of several key components:

- **Java/Kotlin folder:** Contains your code files (activities, logic).
- **res (resources):** Holds XML layouts, images, and strings.
- **AndroidManifest.xml:** Describes app information like permissions and activities.
- **Gradle files:** Manage dependencies and build configurations.

Creating and Running Your First App

The default app already includes one activity — MainActivity. Inside it, the layout file activity_main.xml defines what appears on the screen.

Example code snippet:

xml

 Copy code

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Hello World!"
    android:layout_gravity="center" />
```

MODULE 2: BUILDING AN APP

Introduction

This module teaches how to create functional apps — not just static screens. You'll learn how to add interactivity using buttons, text inputs, and simple Kotlin logic.

Creating the Layout

In `activity_main.xml`, you can design screens by dragging UI elements or coding XML manually. Example:

xml

 Copy code

```
<EditText
    android:id="@+id/nameInput"
    android:hint="Enter your name" />

<Button
    android:id="@+id/submitButton"
    android:text="Submit" />

<TextView
    android:id="@+id/outputText"
    android:text="" />
```



Adding Logic in Kotlin

You connect UI elements to logic in `MainActivity.kt` using **`findViewById`**:

kotlin

 Copy code

```
val button = findViewById<Button>(R.id.submitButton)
val input = findViewById<EditText>(R.id.nameInput)
val output = findViewById<TextView>(R.id.outputText)

button.setOnClickListener {
    val name = input.text.toString()
    output.text = "Welcome, $name!"
}
```

MODULE 3: DISPLAY LISTS AND USE MATERIAL DESIGN

Introduction

Real-world apps show data lists — like contacts or messages. This module covers **RecyclerView**, used to display lists efficiently, and **Material Design**, which gives your app a clean, modern look.

RecyclerView Basics

RecyclerView is a dynamic list view that reuses item layouts for better performance. Steps:

1. Add RecyclerView in layout.
2. Create a list item layout (e.g., list_item.xml).
3. Write an Adapter class to connect data to the list.

Example:

kotlin

 Copy code

```
class MyAdapter(private val data: List<String>) :  
    RecyclerView.Adapter<MyAdapter.ViewHolder>() {  
  
    class ViewHolder(val textView: TextView) : RecyclerView.ViewHolder(textView)  
    override fun onCreateViewHolder(...) = ...  
}
```

Material Design

Material Design provides ready-to-use UI components with attractive animations and consistency. Examples include **Floating Action Buttons (FAB)**, **Cards**, and **Toolbars**.

MODULE 4: NAVIGATION AND APP ARCHITECTURE

Introduction

Modern apps have multiple screens. This module explains how to navigate between them and how to organize your app's structure using architecture components.

Navigation

Navigation can be done via:

- **Intents:** Move from one activity to another.
- **Navigation Component:** A visual tool for controlling app flow.

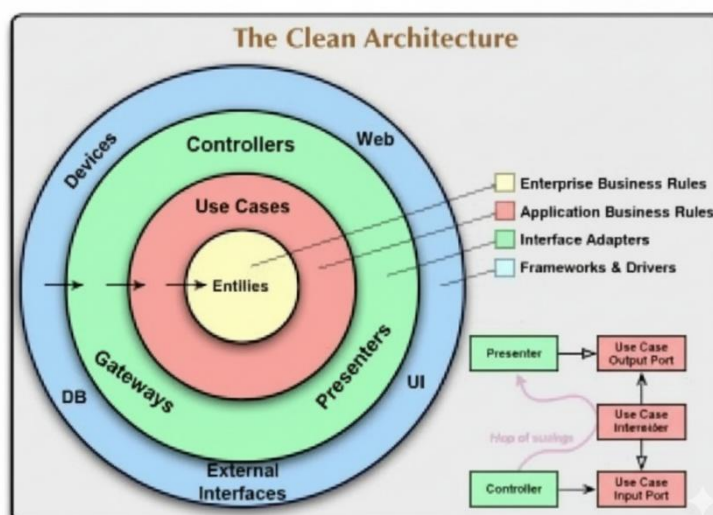
```
val intent = Intent(this, SecondActivity::class.java)
startActivity(intent)
```

Example:

Architecture

Android apps follow the **MVVM (Model-View-ViewModel)** pattern:

- **Model:** Handles data.
- **View:** Displays UI.
- **ViewModel:** Connects data and UI logically.

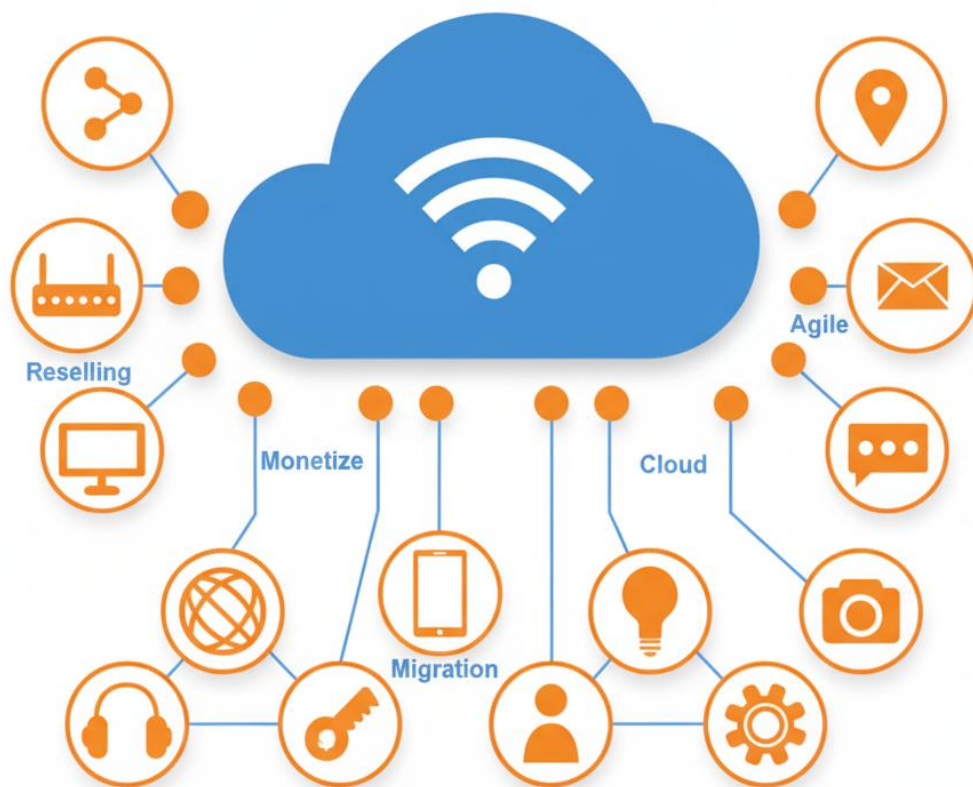


MODULE 5: CONNECT TO THE INTERNET

Introduction

This module focuses on building apps that fetch or send data online, such as weather apps or news feeds.

Connecting to the internet is a cornerstone of contemporary app development, facilitating seamless interaction between client devices and remote servers. Below is a streamlined summary of essential elements involved in enabling internet access within Android applications:



- **Network Permissions:** To allow your app to communicate over the internet, you must specify the required permissions in the AndroidManifest.xml file. This involves adding the `<uses-permission>` tag with the attribute `android:name="android.permission.INTERNET"`. Granting this permission enables your app to open network sockets for data transmission.
- **Making Network Requests:** Applications typically use HTTP or HTTPS protocols to send requests to and receive responses from web servers. Common approaches include:

- **URLConnection:** A core Java API that provides a low-level interface for crafting HTTP requests and handling responses, offering detailed control over the network communication process.
- **HttpClient:** Previously a popular library for HTTP operations, HttpClient has been deprecated in recent Android releases. Developers are encouraged to use alternatives like HttpURLConnection or third-party libraries such as OkHttp for improved performance and support.
- **Reselling (Router icon):** The internet facilitates **reselling** of various services, from web hosting and internet access itself (often through routers and network devices) to digital products and applications. Businesses can white-label or distribute existing services.
- **Monetize (Money/Handshake icon):** Once connected, there are countless ways to **monetize** an online presence. This includes e-commerce, advertising, subscription services, premium content, and leveraging data for targeted marketing.
- **Agile (Swirling arrow icon):** Internet and cloud connectivity enable **agile** development and operations. Teams can collaborate in real-time, deploy updates quickly, and adapt to changes with speed, benefiting from online project management tools and continuous integration/delivery pipelines.
- **Migration (Phone icon with arrow):** The internet, especially the cloud, is crucial for **migration** – moving data, applications, and services from on-premise systems to cloud-based platforms, or between different cloud environments. This often involves mobile devices connecting to these new cloud infrastructures.
- **External Interfaces / Gateways (Globe icon, Key icon):** The internet acts as the primary medium for interacting with **external interfaces** and provides **gateways** for secure access to services. This involves APIs, webhooks, and secure authentication methods (like the key icon suggests) to ensure data integrity and privacy during online interactions.
- **Settings/Configuration (Gear icon):** Managing and configuring connected services often happens over the internet. This icon represents the ability to remotely adjust **settings**, manage user accounts, and control system configurations for cloud-hosted applications or IoT devices.

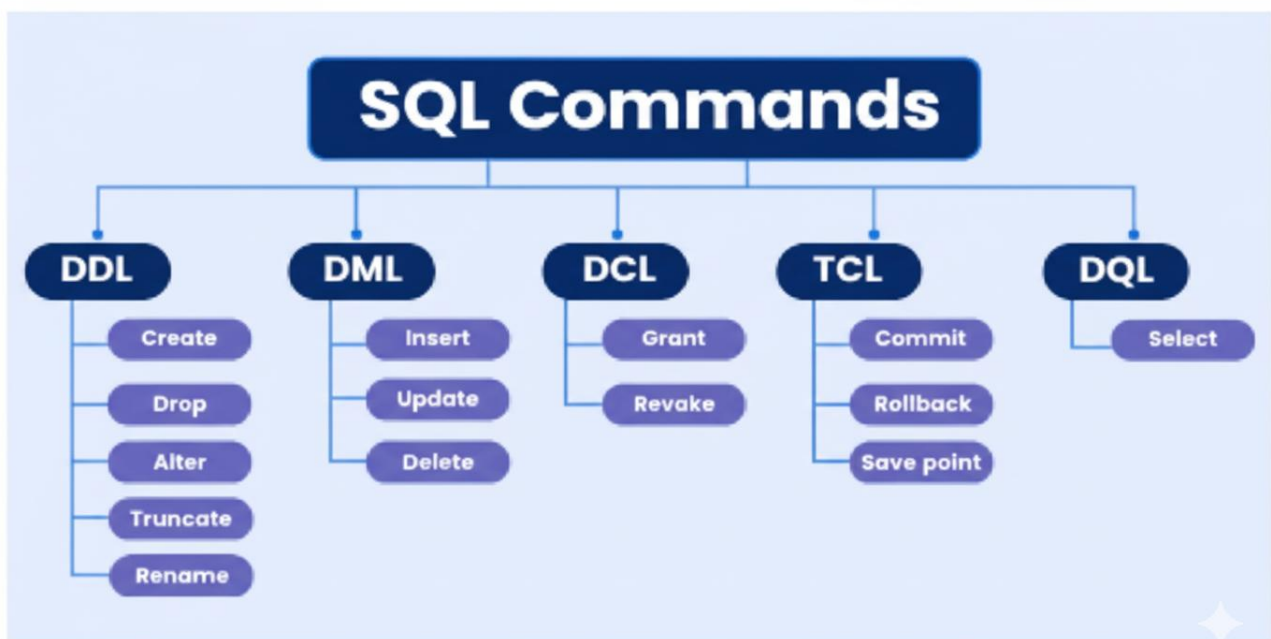
MODULE 6: DATA PERSISTENCE

Introduction

Data persistence is a fundamental concept in software engineering that involves saving data to a stable storage medium—like databases or file systems—and retrieving it when needed. This capability is essential for building reliable and user-friendly applications that maintain state across sessions.

In Android development, there are multiple strategies to achieve data persistence:

- **Structured Query Language (SQL):** SQL is a specialized language crafted for managing relational databases. It offers a standardized method to interact with data by enabling operations such as querying, inserting, updating, and deleting records. Beyond data manipulation, SQL also allows developers to define and modify database schemas, including creating and altering tables, ensuring data is organized and accessible based on specific criteria.
- **DataStore for Key-Value Storage:** Part of the Android Jetpack suite, DataStore provides a modern, efficient way to handle key-value pairs. It serves as a more reliable and type-safe successor to the older SharedPreferences system. DataStore operates by associating each data item with a unique key, simplifying the process of storing, retrieving, and updating values within an app's persistent storage.



MODULE 7: WORK MANAGER

Introduction

WorkManager handles background tasks like syncing or uploading data — even when the app is closed.

- **Robust Background Task Handling:** WorkManager is an essential part of the Android Jetpack library, expertly designed to manage background tasks that operate independently from the app's user interface. This functionality is vital for tasks such as synchronizing data with remote servers, handling file uploads and downloads, and executing routine maintenance processes without disrupting the user's interaction with the app. By offloading these operations to the background, WorkManager not only preserves the app's responsiveness but also optimizes battery and system resource usage, contributing to a smoother user experience.
- **Reliable Task Persistence:** A defining strength of WorkManager lies in its ability to guarantee that scheduled tasks are not lost even if the app is terminated or the device undergoes a reboot. This is accomplished through sophisticated scheduling algorithms and deep integration with the Android operating system, which allow tasks to automatically resume or restart as needed. This persistence is particularly critical for applications that depend on the assured completion of background activities, such as transmitting analytics data, refreshing content at regular intervals, or finalizing lengthy operations that cannot be interrupted.
- **User-Friendly API:** WorkManager simplifies the developer's workload by providing a consistent and easy-to-use API that hides the complexities of Android's diverse background execution frameworks, including JobScheduler, AlarmManager, and Firebase JobDispatcher. This abstraction enables developers to focus on crafting the core logic of their background tasks without worrying about compatibility issues across different Android versions or device manufacturers. Moreover, WorkManager supports advanced features like chaining multiple tasks together and setting flexible execution constraints (e.g., network availability, charging status), empowering developers to build sophisticated and efficient background workflows with minimal effort.

MODULE 8: EXAMPLE PROJECTS

TASK 1 : CALCULATOR APP

The task involved for developing a calculator app. This app provided users with a simple and intuitive interface for performing basic arithmetic operations.



The provided image shows a modern, two-in-one design concept for a basic four-function calculator, offering both a **Light Mode** and a **Dark Mode**.

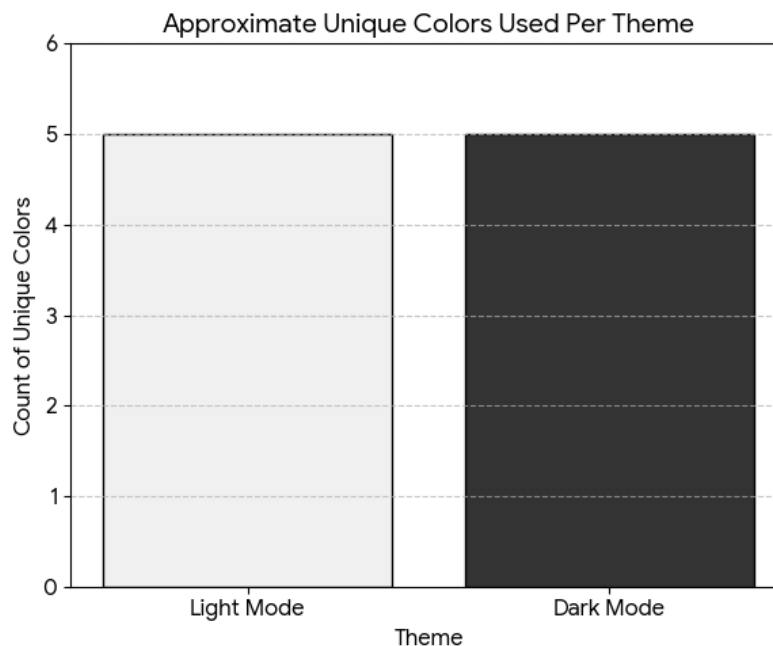
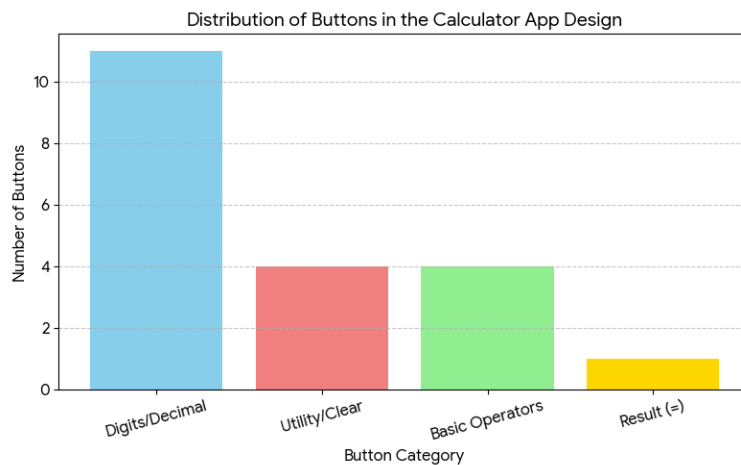
The app adheres to a clean, grid-based layout. It features a large display area at the top for results and input, and a segmented button area below. The button colors are used effectively to categorize functions:

- **Digits and Decimal (11 buttons):** Standard numeral input.

- **Basic Operators (4 buttons):** Addition (+), Subtraction (-), Multiplication (*), and Division (/).
- **Utility/Clear (4 buttons):** Clear (C/AC), sign change (+/-), percentage (%), and delete (backspace).
- **Result (1 button):** Equals sign (=).

The dual-theme design (Light Mode with bright backgrounds and Dark Mode with deep gray/black backgrounds) ensures user comfort and accessibility in different lighting environments.

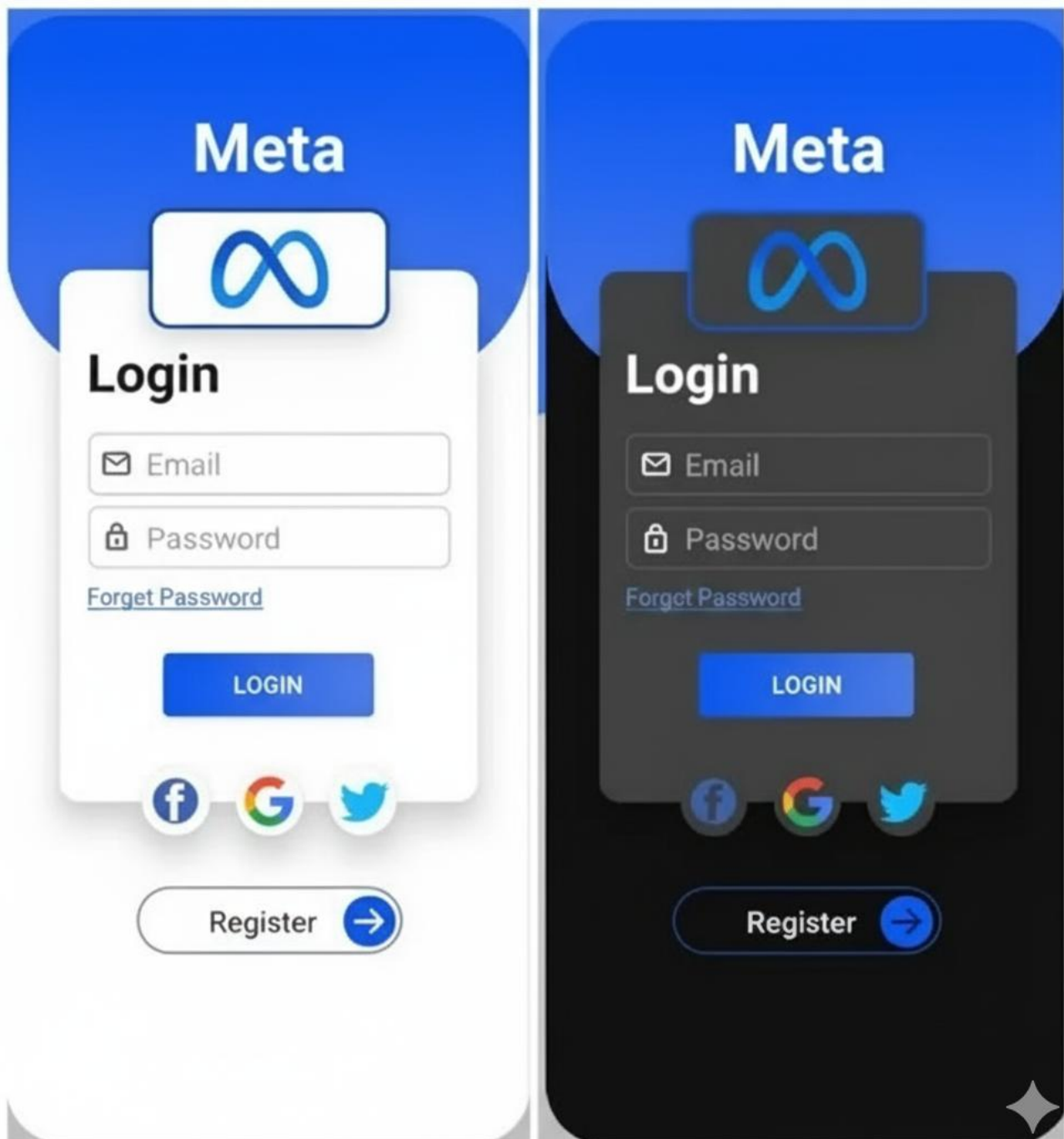
Analysis and Report in Graphs :



Key Observation: The largest category is **Digits/Decimal**, confirming the primary function of the app is numerical input. The balanced presence of Utility/Clear and Basic Operators suggests a focus on quick, common calculations.

TASK 2 : SIMPLE LOGIN INTERFACE

The task involved for developing a user-friendly login interface designed to enhance the security and convenience of accessing your app. With a sleek and intuitive design, our login interface ensures a seamless login experience for users, making it easier than ever to protect their accounts. This modern UI/UX design provides a beautiful interface in both light and dark mode, offering a visually appealing experience for your users



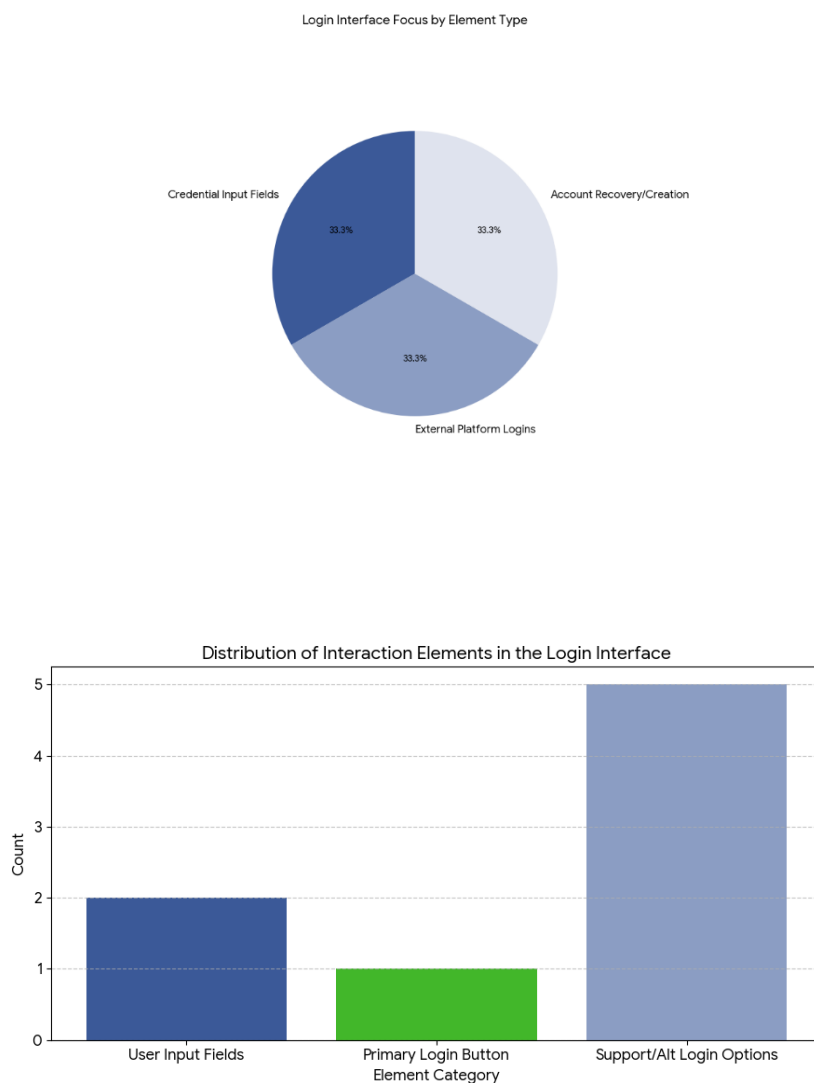
The design prioritizes ease of access by offering multiple authentication pathways:

1. **Standard Credentials:** There are two dedicated input fields for the user (likely Email/Phone Number and Password).

2. **Primary Action:** A clear, high-contrast button is provided for the main login action.
3. **Alternative/Recovery Options:** A significant portion of the screen space is dedicated to options beyond the standard login, including:
 - Links for **Forgot Password** and **Sign Up/Create New Account**.
 - Prominent buttons for **social/alternative logins** (often Facebook and Instagram) to facilitate quick access via pre-existing, trusted accounts.

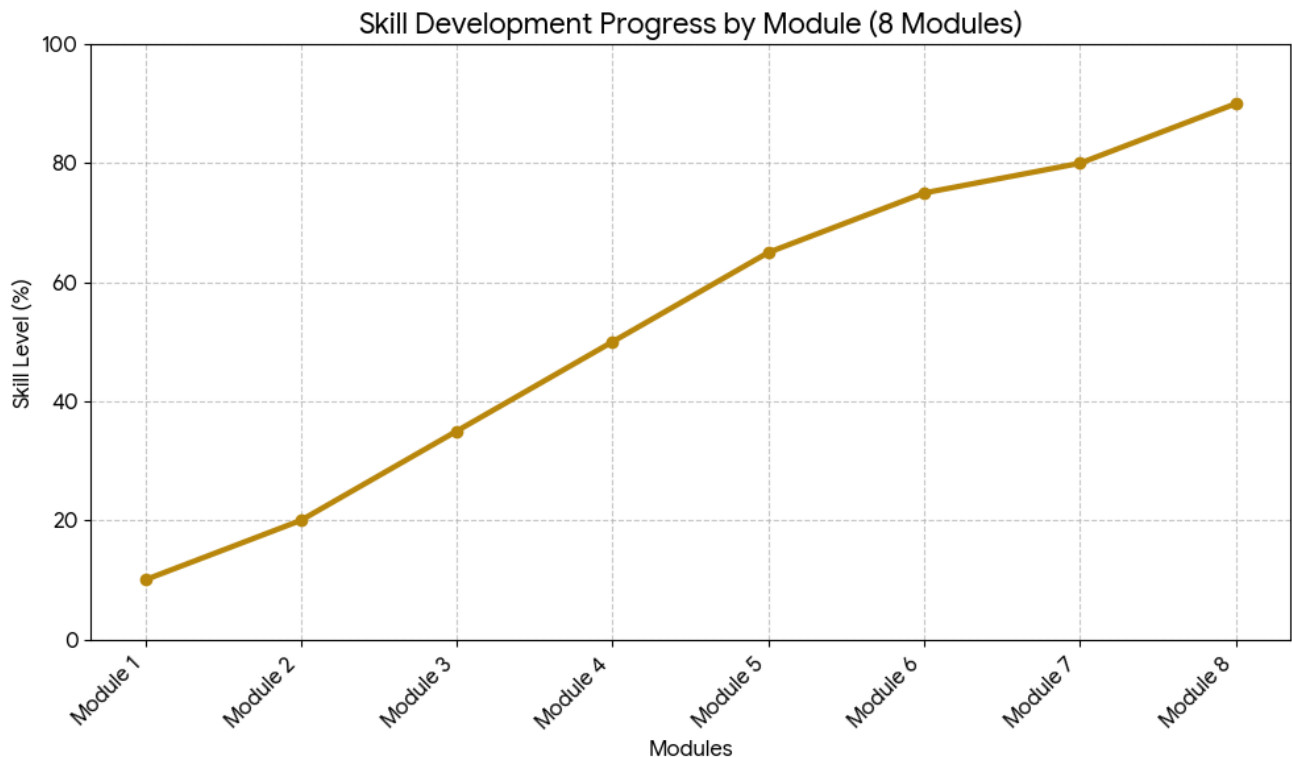
The overall layout is minimalist, centering the login card to focus user attention on the required actions.

Analysis and Report in Graphs :



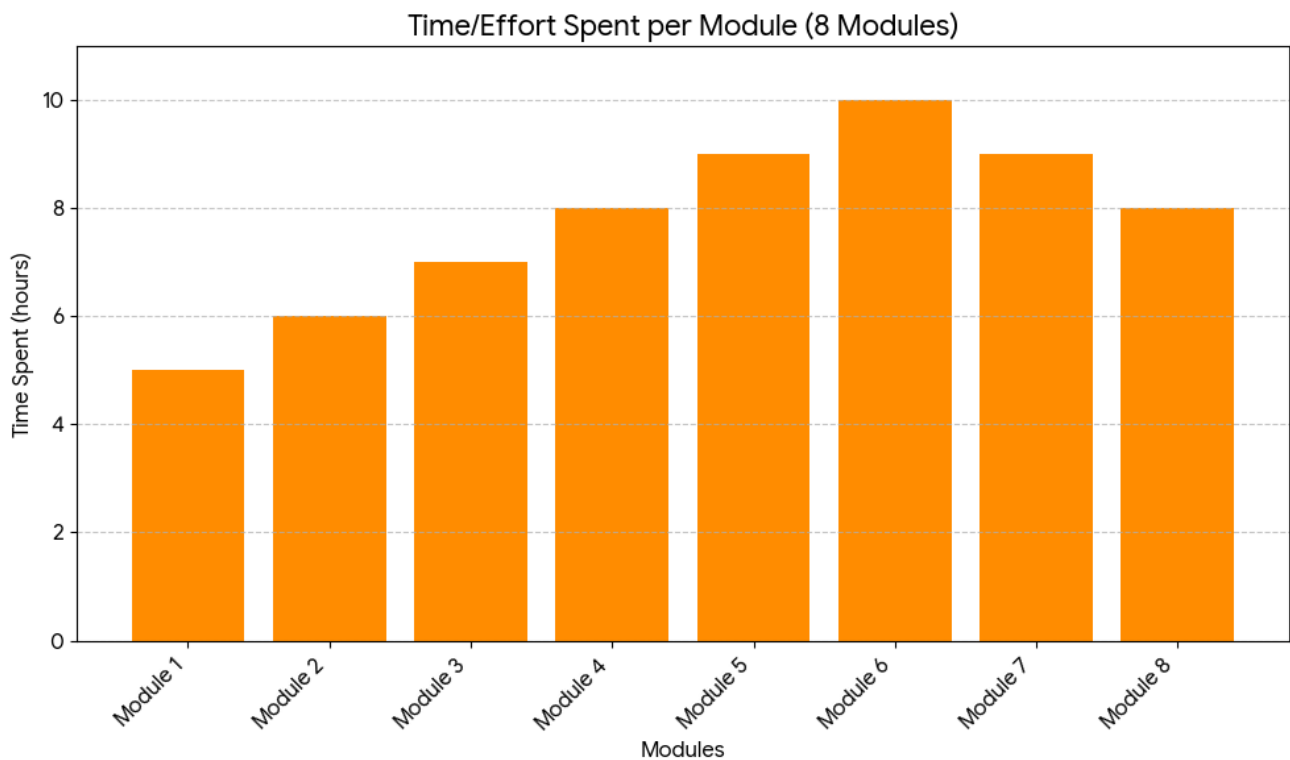
Key Observation: The interface dedicates more resources to **Support/Alternative Login Options** than to the standard input fields and primary button combined. This strategy aims to maximize user success by reducing friction for both existing users (via alt-logins) and users needing recovery or account creation (via support links).

REPORT



This line graph illustrates the cumulative progress in skill level (percentage) across the first eight modules.

- **Continuous Growth:** The line shows a **strong and consistent upward trend**, demonstrating that the intern successfully acquired new knowledge in every module.
- **Reaching High Proficiency:** By the end of Module 8, the intern has reached a high level of competency, achieving **90% skill level**.
- **Learning Curve:** The steepest part of the curve occurs in the early modules, where knowledge gain is most rapid (e.g., jumping from 10% to 50% by Module 4). The growth rate slows slightly in the later modules (Modules 6-8) as the learner moves from basic understanding to advanced, specialized concepts (75% to 90%), which is typical when approaching mastery.



This bar chart displays the time commitment (in hours) required for the first eight modules. The data shows a progressive increase in effort, forming the first part of a classic **bell-shaped curve**:

- **Initial Effort (Modules 1-4):** The time commitment steadily increases from a low of 5 hours in Module 1 to 8 hours in Module 4. This reflects the increasing complexity of the foundational topics.
- **Peak Challenge (Modules 5-7):** The effort plateaus at its highest point in the middle section, with **Module 6 requiring the maximum 10 hours**. This indicates that the core topics covered in these modules (likely relating to networking, architecture, and data persistence) were the most challenging and time-consuming.
- **Slight Decrease (Module 8):** The time spent drops back to 8 hours in Module 8, suggesting that the intern began to apply the skills from the peak modules more efficiently, or the complexity of that specific module was slightly lower.

CONCLUSION

The Google Android Development internship provided a comprehensive learning experience in building Android applications, covering essential aspects from creating the first app to incorporating advanced features like data persistence and background processing. Throughout the internship, I gained proficiency in developing user interfaces following Material Design principles, enabling me to create visually appealing and intuitive apps. Additionally, the emphasis on displaying lists efficiently enhanced my ability to present data effectively within applications. Starting with the creation of their first Android app, interns progressed through building intuitive user interfaces, implementing Material Design principles, structuring app navigation and architecture, integrating internet connectivity, managing data persistence, and mastering background task processing.

The internship provided a structured curriculum that equipped interns with essential skills and knowledge, ensuring they emerged proficient in developing professional-grade Android applications. From foundational concepts to advanced topics, interns gained valuable hands-on experience and emerged ready to embark on successful careers as Android developers.

Additionally, the internship fostered a collaborative learning environment where participants engaged in practical exercises, code reviews, and projects, allowing them to apply theoretical knowledge to real-world scenarios. Through mentorship and peer interaction, interns not only honed their technical skills but also developed critical problem-solving abilities and effective communication within a team setting. Furthermore, the internship encouraged continuous learning and adaptation to the ever-evolving landscape of Android development, instilling in interns the mindset of lifelong learners equipped to tackle new challenges and innovations in the field.

In conclusion, the Google Android Development internship provided a wellrounded education in Android app development, covering fundamental concepts as well as advanced techniques. The hands-on experience gained throughout the program has equipped me with the skills necessary to create impactful and featurerich Android applications